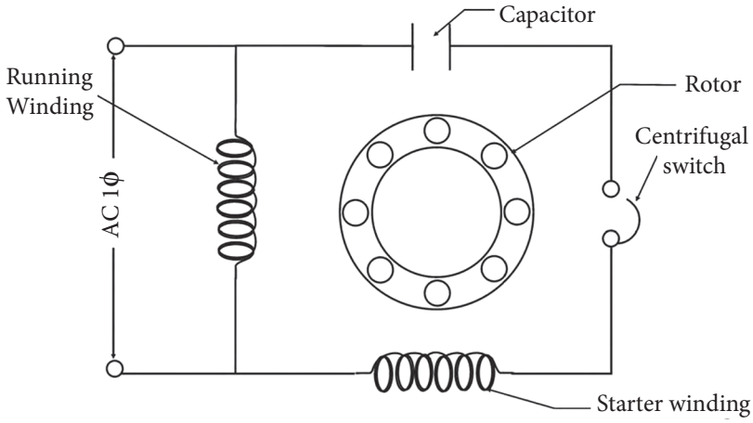


A.C Single Phase Capacitor Start Motor

A.C Single Phase Capacitor Start Motor

The stator of a A.C single phase capacitor start motor has two windings- running winding & starting winding. A capacitor and centrifugal switch are connected in series with the starting winding. A squirrel cage rotor is fitted in this motor. Capacitor creates 90° phase difference between the two windings.

Single phase current is supplied and the development of rotating magnetic field makes the rotor to rotate. When the rotor attains 70% of the total speed, the centrifugal switch disconnects the starting winding, this motor has high starting torque. The direction of rotation can be changed by just changing the connection of terminals of anyone of the windings. The motors find extensive use in wet grinders, small grinding machines, drilling machines, compressed motors used for air-conditioners and refrigerators.

10.8 STARTERS FOR INDUCTION MOTORS

Need of Starters

When induction motors are directly switched on to supply, it takes about five to six times of full load current. This initial excessive current causes damages to the motor and supply wires. Starters are used to limit the inrush of starting line current and full current is supplied when the motor picks up speed. It consists of

DO YOU KNOW?



Georg Simon Ohm (16 March 1789 – 6 July 1854) was a German physicist and mathematician. As a school teacher, Ohm began his research with the new electrochemical cell, invented by Italian scientist Alessandro Volta. Using equipment of his own creation, Ohm found that there is a direct proportionality between the potential difference (voltage) applied across a conductor and the resultant electric current.

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